WANYANG YU

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EDUCATION

University of Science and Technology of China Sep 23' – Jun 27' (Exp.)
B.E. in Computer Science and Technology *Hefei, China*
Hua Xia Talent Program | GPA: 3.86/4.3 | Weighted average: 89.76/100 | Rank: 23/236
Transferred from the Division of Life Sciences and Medicine in 2024.

RESEARCH EXPERIENCE

Cognitive Modeling of Scientific Inquiry / Visiting Researcher Oct '26 – Present
Computational Cognitive Science Lab, School of Informatics, University of Edinburgh *Edinburgh, UK*
Advisor: Dr. Bonan Zhao
Technical areas: Computational modeling, NLP, cognitive science.

Interface for Electrical Stimulation / Summer Research Intern Jul '26 – Present
Human Computer Integration Lab, University of Chicago *Chicago, USA*
Advisor: Prof. Pedro Lopes | PhD mentor: Yun Ho
Technical areas: Electrical stimulation, interface prototyping, software development.

Longitudinal Brain MRI Prediction / Undergraduate Researcher Oct 25' – Feb 26'
National Engineering Lab for Brain-inspired Intelligence Technology, USTC *Hefei, China*
Advisor: Prof. Aiping Liu

- Developed DRIFT, a two-stage deep learning framework to predict follow-up 3D brain MRIs from baseline scans and elapsed time, using a pretrained brain-age model and a conditional 3D U-Net.
- Modeled anatomical change through diffeomorphic deformation fields with bounded intensity correction; produced velocity fields and Jacobian maps for inspection of predicted changes.
- Independently led problem formulation, model design, implementation, experiments, and manuscript writing; evaluated on 3,926 UK Biobank longitudinal MRI pairs.

mmWave Backscatter Tags for Drone Identification / Undergraduate Researcher Jun 25' – Jan 26'
Lab for Intelligent Networking and Knowledge Engineering, USTC *Hefei, China*
Advisor: Prof. Yubo Yan

- Designed and simulated backscatter tags for a mmWave communication system, using electromagnetic simulation and PCB design to investigate signal reflection and tag identification.
- Investigated discrepancies between an open-source design and prototype performance; initiated technical exchanges with a Peking University PhD researcher encountering the same issue.

ENGINEERING PROJECTS

LLM Assisted Memory Management / Group Leader — Mar – Jun 25'
USTC course project/ Operating Systems (H) — *Hefei, China*
Instructor: Prof. Kai Xing

- Led a 4-person team developing a predictive memory pre-allocation framework using locally deployed Llama 3-8B-Instruct to anticipate user actions in web tasks.[Code]

VEGFR-like Surface Display on E. coli for Tumor Inhibition / Student Leader — Jan – Oct 24'
USTC team / International Genetically Engineered Machine Competition — *Hefei, China*
Advisor: Prof. Jiong Hong

- Contributed to experimental design and data analysis for VEGFR-like surface display on E. coli
- Led scientific communication and the team presentation at the Grand Jamboree in Paris.

AWARDS

UChicago / Computer Science Summer Research Program Fellowship	2026
USTC / Excellent Student Scholarship	2025, 2026
USTC / Talent Program Scholarship in Computer Science and Technology	2024, 2026
USTC / Rose Scholarship (Top 1%)	2024
USTC / Alumni Foundation High School Scholarship Award	2023

SKILLS

Programming: Python, C/C++, MATLAB, Verilog HDL
Machine learning and vision: PyTorch, MindSpore, MediaPipe, GVHMR; 3D medical image modeling
Hardware and simulation: Electrical muscle stimulation, KiCad, Ansys HFSS, Advanced Design System
Research tools: Git/GitHub, LaTeX; experimental design and analysis workflows
Languages: English (TOEFL 104), Mandarin (Native)

SELECTED COURSEWORK

Brain–Computer Interface (95) · Operating Systems (H) (95) · Stochastic Processes B (92) · Probability and Mathematical Statistics (91) · Intelligent Computing Software Optimization (95)

LEADERSHIP AND COMMUNITY

USTC Literary Society / President — Jul 25' – Jun 26'
Organized meetings and reading groups; designed posters and coordinated the society's literary magazine.

Hometown Community Fieldwork / Volunteer — Jan 25'
Interviewed retired factory workers and a supercomputing center director, and wrote articles documenting their experiences and perspectives.

Science Outreach / Volunteer — Jun 24'
Introduced children on the autism spectrum to *E. coli* through a shared drawing activity.